MINING DEALER BEST PRACTICE

Tool for Installing 3500 Camshaft Gears

Maintenance and Repair | Application | Component Life Management | **Component Renewal (CRC)** | MARC Management

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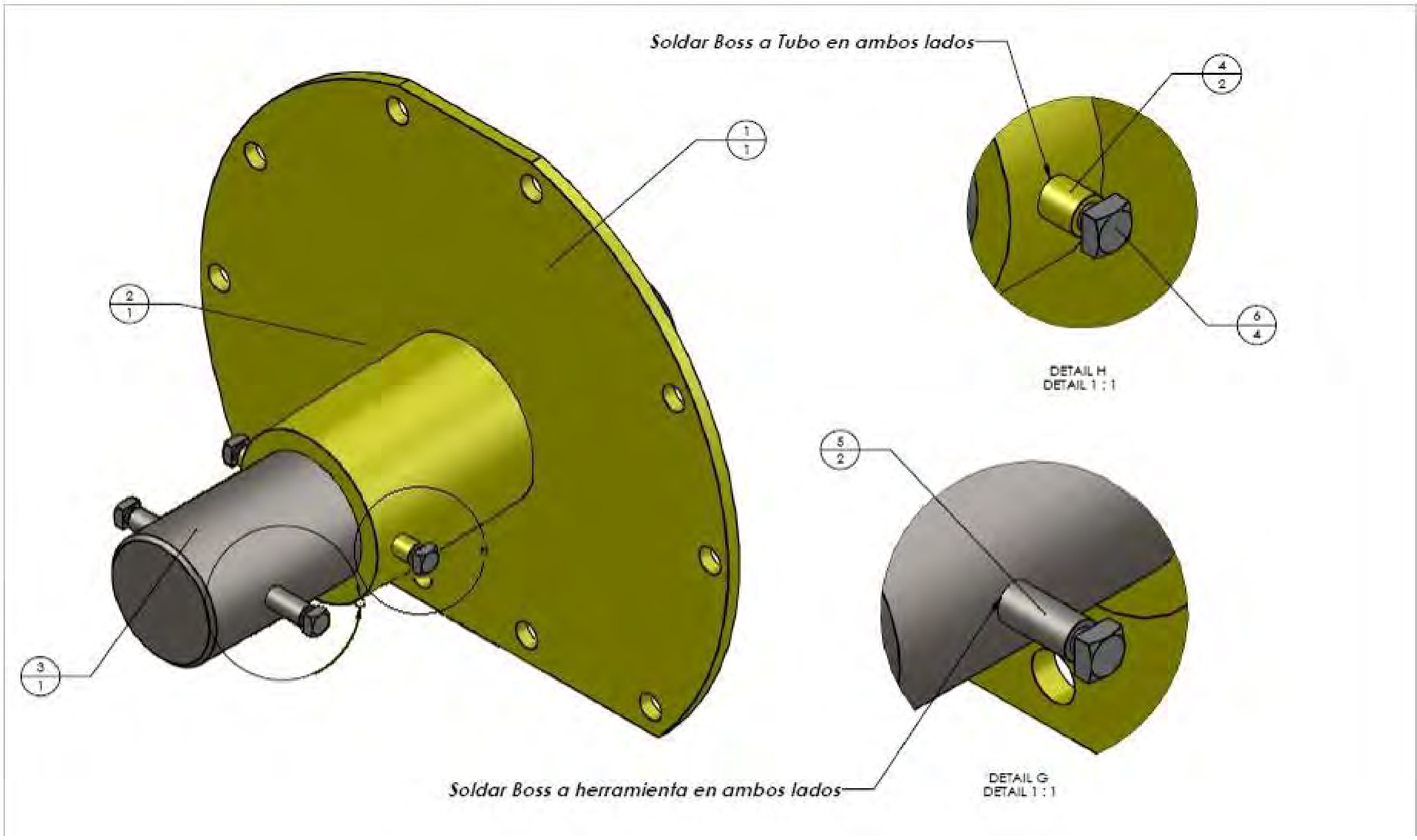
1.0 Introduction

There is not a Caterpillar tool for the installation of camshaft gears in 3500 engines. Technicians often use a variety of different manual methods. Most of the manual methods are inefficient and handling of the tools is difficult.

2.0 Best Practice Description

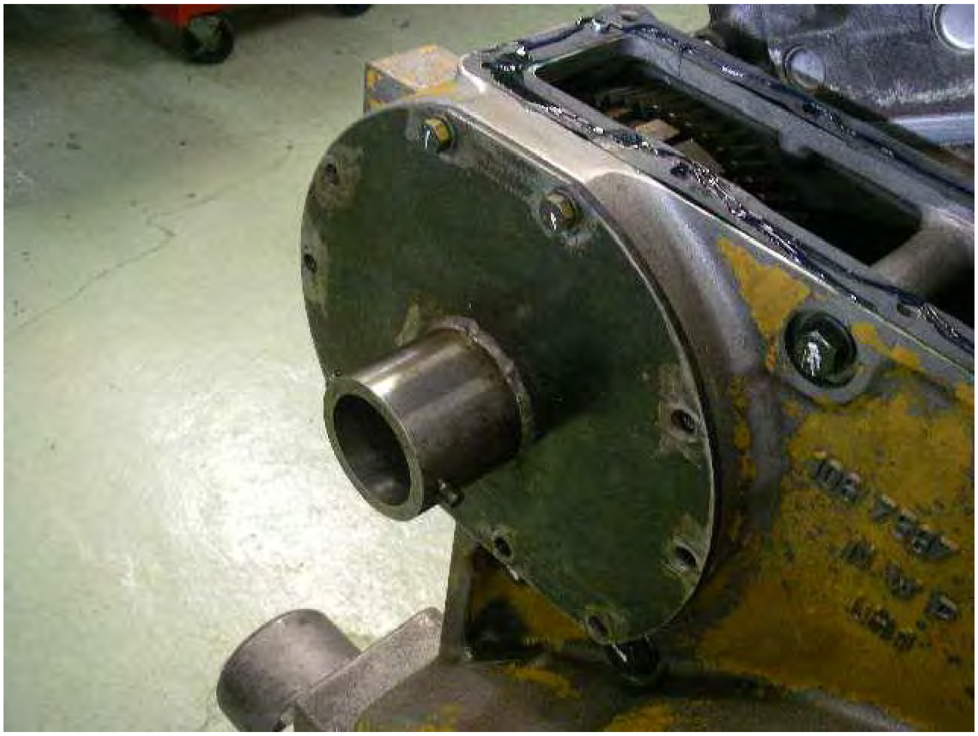


Gecolsa has developed a tool to facilitate the assembling of camshaft gears in 3500 engines. The tool consists of

- plate support (1) to hold the tool with the work piece
- Tube (2) to guide the bar
- Preloading bar (3)
- Four bosses for the screw (4 and 5)
- Four screws (6)
- Elastic ring (7)

3.0 Implementation Step

First the tool is positioned on the engine block as is shown in the picture.



The second step is introducing the bar in the guide tube.

The third step is holding the tool with the screws and the elastic rings.

Now technicians can apply the force in a safe way.

Finally torque is applied with a extension

4.0 Benefits

- The tool makes the installation process of the camshaft gears safer.
- The tool increases the efficiency of the process, requiring less repetitions to finish the work.
- The tool increases the quality of the process; the technicians can apply precise and centered forces.

5.0 Resources Required

Component List					
Part	Name	Description	Amount	Component size	Material
1	Platen	1/4"	1	250,100 mm x 250,100 mm	A -36
2	Tube	Dia Int 60,250 mm x Dia Ext 79,10 mm	1	100 mm	AISI 4130
3	Bar	Dia 60,050 mm	1	210 mm	AISI 4130
4	BOSS	Dia 3/8"	2	19,23 mm	AISI 4130
5	BOSS	Dia 3/8"	2	9,7 mm	AISI 4130
6	Screw	Dia 1/4" NC x 1" Long	4		A -36
7	Elastic ring	PN: 5P-7815	2		

6.0 Supporting Attachments / References

- Manufacturing Drawings.pdf

7.0 Related Best Practices

8.0 Acknowledgements

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